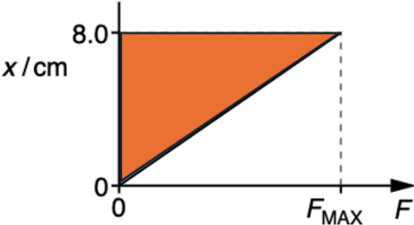
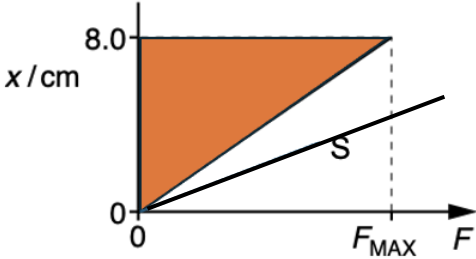


7	(a)	(i)	$m_x u_x + m_Y u_Y = m_x v_x + m_Y v_Y$ $50(4.5) + M(-2.8) = 50(-1.8) + M(1.4)$ $M = 75 \text{ g}$	M1 A1
		(ii)	<u>Total kinetic energy of the system before and after collision is the same.</u>	B1

		(iii) Using KE: $\frac{1}{2}m_x u_x^2 + \frac{1}{2}m_y u_y^2 = \frac{1}{2}[0.050(4.5)^2 + 0.075(-2.8)^2] = 0.80 \text{ J}$ $\frac{1}{2}m_x v_x^2 + \frac{1}{2}m_y v_y^2 = \frac{1}{2}[0.050(-1.8)^2 + 0.075(1.4)^2] = 0.15 \text{ J}$ Since final KE < initial KE, inelastic collision OR Using relative speed $u_x - u_y = 4.5 - (-2.8) = 7.3 \text{ m s}^{-1}$ $v_y - v_x = 1.4 - (-1.8) = 3.2 \text{ m s}^{-1}$ Since relative speeds are not the same, inelastic collision	M0 M1 A1 (M0) (M1) (A1)
		(iv) Since <u>total initial momentum is not zero</u>, having <u>both objects concurrently at rest means that instantaneous total momentum is zero</u>. Hence, not possible due to conservation of linear momentum	B1 B1
	(b)	(i) 	B1
		(ii) $\Delta KE = \Delta EPE$ $\frac{1}{2}mv^2 = \frac{1}{2}F_{MAX}x$ $\frac{1}{2}(0.50)(0.25)^2 = \frac{1}{2}F_{MAX}(0.080)$ $F_{MAX} = 0.39 \text{ N}$	M1 A1
		(iii)  B1: Line should be less steep B1: Line should extend further rightward than original	B1 B1
	(c)	(i) Net increase in weight when ballasts are completely filled with sea water from 25% full: $\Delta W = 0.75(2 \times 50)(1030)(9.81)$ $\Delta W = 757822.5 = 7.6 \times 10^5 \text{ N}$	M1 A0

		(ii) New mass of submarine $m = 2.2 \times 10^6 + \frac{757822.5}{9.81} = 2277250 \text{ kg}$ $a = \frac{\Delta W}{m} = \frac{757822.5}{2277250}$ $a = 0.33 \text{ m s}^{-2}$	M1 A1
		(iii) When the submarine descends at an angle, the <u>surface area that is considered for drag is smaller.</u> Hence <u>the downward component of velocity of the submarine is higher</u> when it is descending at an angle.	M1 A1
		(iv) The <u>water molecules gain an upward momentum</u> after impact with the angled dive plane. Hence by <u>N2L</u>, an <u>upward force is exerted on the water molecules due to the dive plane.</u> By <u>N3L</u>, a <u>downward force is exerted on the dive plane by the water molecules,</u> resulting in the submarine being pushed downwards	M1 A1 B1

2	(c)	Binding energy = [Total mass of the unbound nucleons - mass of iron nucleus] c^2	B1
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